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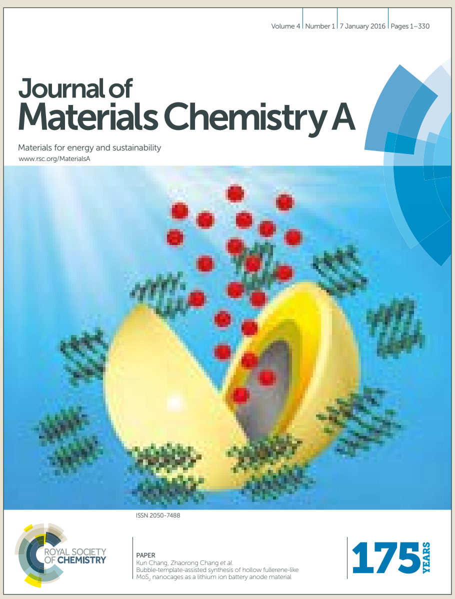

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PAPER

Enhanced cycling stability of boron-doped lithium-rich layered oxide cathode materials by suppressing transition metal migration

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Lithium-rich layered oxide (LLO) has been considered as an attractive candidate due to its high capacitive performance. However, its practical applications are hindered by voltage/capacity fading, low initial Coulombic efficiency, and nonnegligible Mn^{2+} dissolution. Herein, we have prepared boron-doped LLO (LLO@LBO) with excellent cycle/voltage retention and improved initial Coulombic efficiency by a facile synthesis process. The first-principle calculation and ex-suit XRD have clarified that the incorporation of boron atoms in the tetrahedral interstice can effectively impedes the migration channel of the transition metal ions. The strong B-O bonds can enhance the structural stability of LLO@LBO. Furthermore, the boron doping technique can suppress Mn^{2+} dissolution in LLO@LBO during cycling and improve the life time of cathode and anode electrode simultaneously. In the lithium ion battery cathode tests, the LLO@LBO delivers a reversible capacity of 293.9 mA h g^{-1} with an capacity retention of 89.5% at 0.5 C after 100 cycles. While the full cell tests shows an initial energy density of 472.1 Wh kg^{-1} with an excellent energy retention of 84.1% after 150 cycles.

Introduction

Lithium-ion batteries (LIBs) have been proved to be one of the most promising energy storage devices to power the electric vehicles (EVs), and hybrid electric vehicles (HEVs) due to their high energy density and long cycle life.[1] Nevertheless, the dissatisfactory energy density of LIBs still limits their driving ranges and practical applications.[2] It is generally believed that the capacity of the cathode materials is one of the major factors of the performance of LIBs.[3,4] With regard to this viewpoint, LLO materials such as $Li_{1+x}TM_{1-x}O_2$ (TM = transition metal Mn, Ni or Co) has become a possible future choice because of their high specific capacity and low cost.[5-7] However, LLOs suffer from fundamental problems including serious voltage/capacity fading, low initial Coulombic efficiency and nonnegligible Mn^{2+} dissolution to some extent.[8,9] Great efforts have been made to settle these problems.[10-14]

Doping of foreign ions is a common strategy to suppress the phase transformation and mitigate the voltage fading. For example, monovalent metal ions,[11,15] multivalent cations,[16,17] F^- anions,[18] polyanions,[19] *etc.* can be incorporated into the lattice sites of the cathode materials. Monovalent metal ions usually have greater atom radii, thus only a small amount of Na^+ and K^+ is enough to increase the slab space of Li^+ and also accelerate the Li^+ transport during cycling.[10,11] Multivalent cations, such as Al^{3+} and Ti^{4+}, can be doped by replacing the intrinsic transition metal ions.[16,20] Moreover, foreign multivalent cations can block the TM migration path owing to their greater atom radii or enhance the structural stability by forming stronger bonds with oxygen.[17] Simple doping of F^- anion has also been extensively studied that it can suppress the formation of oxygen vacancies in the first charging process.[21,22] Besides, F^- cannot provide equal negative charges compared to O^{2-}, which may induce the partial reduction of TMs and provide extra capacity.[18,23] Polyanion doping, including PO_4^{3-},[19] SO_4^{2-}, SiO_4^{4-},[24] and BO_3^{3-},[25] can lower the energy of the O 2p band compared to the pristine layered oxides and limit the evolution of oxygen.[26] The strong covalent bonding of X-O (X=P, S, Si, and B) can improve the thermal and structural stability obviously. However, incorporation of such inactive atoms usually results in lowered specific capacity.[27]

Among them, boron atoms have the lightest atom mass which can reduce the adverse impact from the incorporation of inactive atoms to some extent. As we know, TM ions can migrate from octahedral sites of the TM layers to tetrahedral sites of the Li layer, then further migrate to octahedral sites of the Li layer.[10] Due to the minor radius of boron (0.27 Å), it can be intercalated into the tetrahedral sites of the TM layer.[28] Recently, it has been demonstrated that the introduction of boron atoms can impede the path and then prevent the phase transformation from layer structure to spinel structure.[25,28,29] However, the electrochemical performance enhancing mechanism of boron doping is still elusive. In addition, the doping methods previously used, such as sol-gel process require rather complex synthetic routes which limits its practical application.[30-32] Besides, the full cell performance is needed to evaluate reliably the material performance, which is still missing.

Herein, we present a boron-doped LLOs with a facile preparation method using $LiBO_2$ as the boron source. The

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structural and electrochemical performance of LLOs with and without boron doping were compared comprehensively. The electrochemical performance enhancing mechanism of boron is systematically investigated and understood using both experimental and theoretical approaches, including the first-principle calculation and ex-suit XRD (X-ray diffraction). The results indicate that strong B-O bonds can enhance the structural stability of LLO@LBO, and suppress Mn migration and dissolution in LLO@LBO for the life time. With these, the doped material LLO@LBO delivers a reversible capacity of 293.9 mA h g^{-1} with a capacity retention of 89.5% at 0.5 C after 100 cycles. Furthermore, when combined with an anode material SnO_2, the full battery cell shows a high initial energy density (472.1 Wh kg^{-1}) with excellent energy retention (84.1% retention after 150 cycles at 0.5 C).

Result and Discussion

Structure and morphology characterization

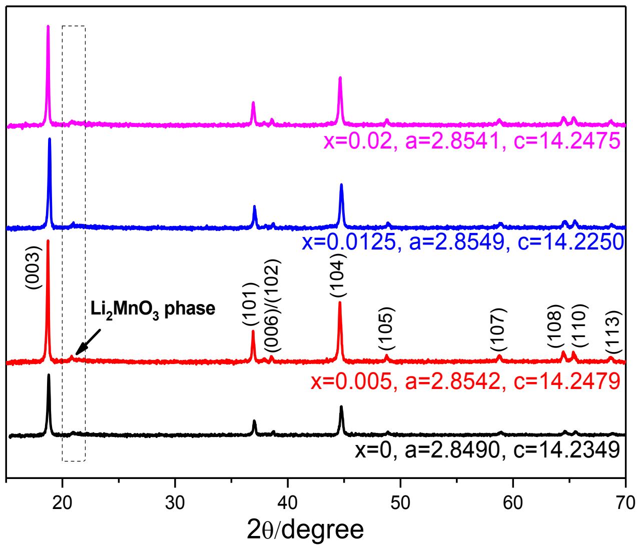

Fig. 1 XRD patterns of Li1.2Ni0.13Co0.13Mn0.54BxO2 (x=0, 0.005, 0.0125 and 0.02)

We obtained the boron doping LLO by a facile solid reaction at high temperature using $LiBO_2$ as boron source. The detailed prepared process is shown in the Experimental section. Figure 1 shows the XRD patterns of $Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}B_xO_2$ (x=0, 0.005, 0.0125 and 0.02). All peaks are indexed on the basis of α-$NaFeO_2$ structure with space group R-3m of layered $LiTMO_2$ (TM =Ni, Co, and Mn).[33] The weak, broad peaks at 21-23° are contributed to the $LiMn_6$ superstructure of the monoclinic Li_2MnO_3 phase with C2/m symmetry. A high (003)/(104) peak intensity ratio (>1.2) were observed clearly, suggesting a low degree of TM ions in the Li layers.[34] The clear peak separation between (006)/(102) and (108)/(110) demonstrates a well-crystallized layered structure.[35] In the XRD patterns of boron-doped compound, no extra peak compared to that without B doping indicates that the boron atoms are incorporated into the lattice of $Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}B_xO_2$. Furthermore, Doping LLOs with boron does not changes its intrinsic layered structure. The unit cell parameters (*a* and *c*) of bare LLOs and B-doping LLOs have no visible difference. Besides, the XRD results have been refined to further study the effect of boron doping on the lattice structure (Table S1 and Figure S1, ESI†). By comparation of refinement results of LLO with and without boron doping, it is concluded that The position of oxygen atoms changes slightly influenced by doped boron atoms. Besides, the Li/Ni mixing increases slightly after boron doping, which is possible induced by high treatment temperature (700 ℃) in the doping process. The atomic occupancies for 3a and 6c sites show unchanged. These results can be explained as the boron atoms are located in crystallographic sites instead of the constituent atoms in LLOs.[36]

To explore the effect of boron doping on the morphology and structure of $Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}B_xO_2$ (LLO@LBO, taking x=0.0125 as an example), the as-prepared samples were characterized by SEM (scanning electron microscopic), TEM (transmission electron microscopy) and XPS (X-ray photoelectron spectroscopy). The spherical LLO@LBO cathode particles shows a radius range of 3~5 μm and are different from the brick-like morphology of LLO (Figure S2, ESI†). Moreover, the primary particles turn to be irregular after boron doping (Figure S2, ESI†). This results from the ball-milling and calcination during the preparation process. The TEM images of LLO and LLO@LBO (individual particle) have been shown in the Figure S3.There are some impurities at the surface of LLO@LBO contributed to residual oxidation production of $LiBO_2$. Figure 2a shows the surface structure of LLO@LBO from TEM. We can easily find two different sets of lattice fringe, corresponding to (003) planes of the R-3m structure and (200) planes of C2/m structure both related to layered LLO@LBO. These are consistent with the XRD results (Figure 1). The TEM-EDS (Energy Dispersive Spectrum) mapping (Figure 2b-f) demonstrate that the distribution of Mn, Ni, Co, and O is uniform, indicating that the introduction of boron atoms has little effect on the distribution of TM. The elemental composition of the surface and bulk LLO@LBO are further analysed by Ar^+ etching. Figure 2g shows the B 1s spectra of LLO@LBO at different etching depth (0, 25, 50, 75, and 100 nm). A sharp peak at around 191 eV was observed which corresponds to the presence of boron atoms.[36] However, the observed that the B 1s spectra at surface (0 nm) shifted to the high bonding energy region (191.8 eV) compared to that at other depths. It is also O 1s spectra (Figure 2h) has the similar tendency. To explain this in detailed, the XPS characterization of B 1s in $LiBO_2$ and its possible oxidation production (B_2O_3) are supplied in the Figure S4. it is obvious that the B 1s peak position (191.8 eV) of LLO@LBO at surface is different from that of $LiBO_2$ (194.2 eV). This means the different chemical environment for boron atoms in LLO@LBO and $LiBO_2$. The B 1s in B_2O_3 is around 192 eV, slightly higher than that in LLO@LBO at surface. Thus, it is possible that the residual oxidation production of $LiBO_2$ induces the B 1s peak shift to higher bond energy region (Figure S3 and S4, ESI†). Consideration of the obvious difference between B 1s in the bulk of LLO@LBO (around 191 eV) and B_2O_3 (192 eV), it is believed that these boron atoms in bulk of LLO@LBO shows different chemical environment with latter.[37] Importantly, the

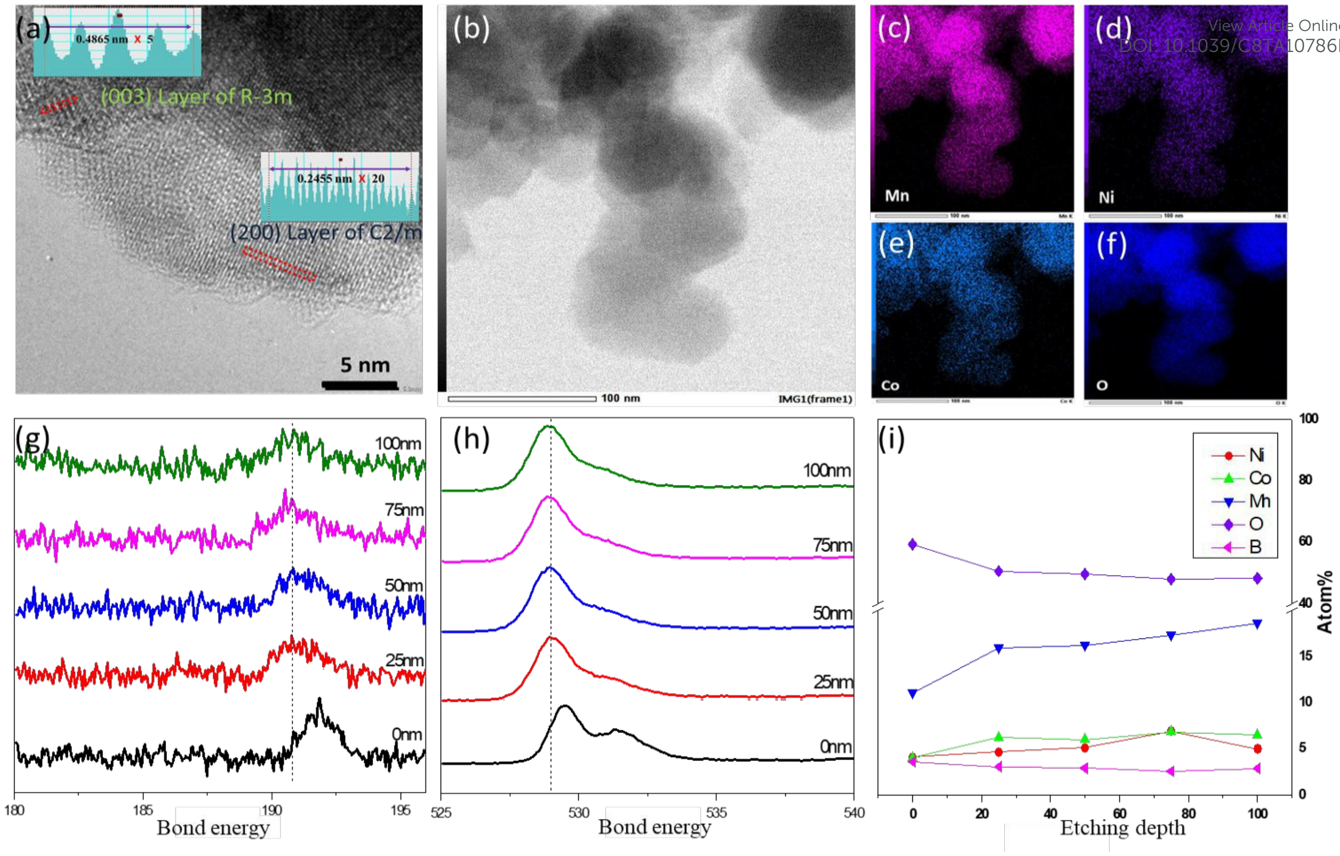

Fig. 2 (a) The HR-TEM pattern of LLO@LBO, (b) the morphology of LLO@LBO and its EDS mapping of (c) Mn, (d) Ni, (e) Co, and (f) O. XPS spectrums of (g) B 1s and (h) O 1s at different etching depth (0 nm, 25 nm, 50 nm, 75 nm and 100 nm). (i) Content of Ni, Co, Mn and B at different etching depths.

concentration of boron shows no meaningful change in bulk structure while that at surface region shows a slightly increasement likely due to the residual oxidation production of $LiBO_2$ (Figure 2i and S3, ESI †). Moreover, the valence states of transition metal (Ni, Co and Mn) remains the same before and after boron doping, supported by XPS results (Figure S5, ESI†).

The FT-IR is conducted as the supplement of XPS test to further study the variation of M-O bond after boron doping (Figure S6, ESI†). The characteristic peaks at 622.97 and 545.82 cm^{-1} are attributed to the O-M-O bending and the M-O asymmetric stretching modes in MO_6 octahedra, respectively. After boron doping, the bending and stretching mode frequencies shift slightly to lower wavenumber regions, indicating that the M-O covalency in LLOs is decreased after boron doping. This is favourable for stabilization of the host lattice in LLOs.[38] Otherwise, the extra peak at 867.91 cm^{-1} is found for LLO@LBO, which is contributed to the asymmetric stretching mode of four-coordinated BO4.[39]

These results indicate that the boron atoms have been successfully doped into LLO with a uniform distribution and the crystal structure and transition metal valence remain unchanged.

Electrochemical results and discussion

A series of materials with different B doping amounts and preparation conditions have been investigated (Figure S7, ESI†) and the optimized material was obtained with the ratio of $Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}B_xO_2$ with x = 0.0125 prepared at the reaction temperature of 700 ℃. Unless otherwise specified, this material, short named as LLO@LBO, was carried out for further studies.

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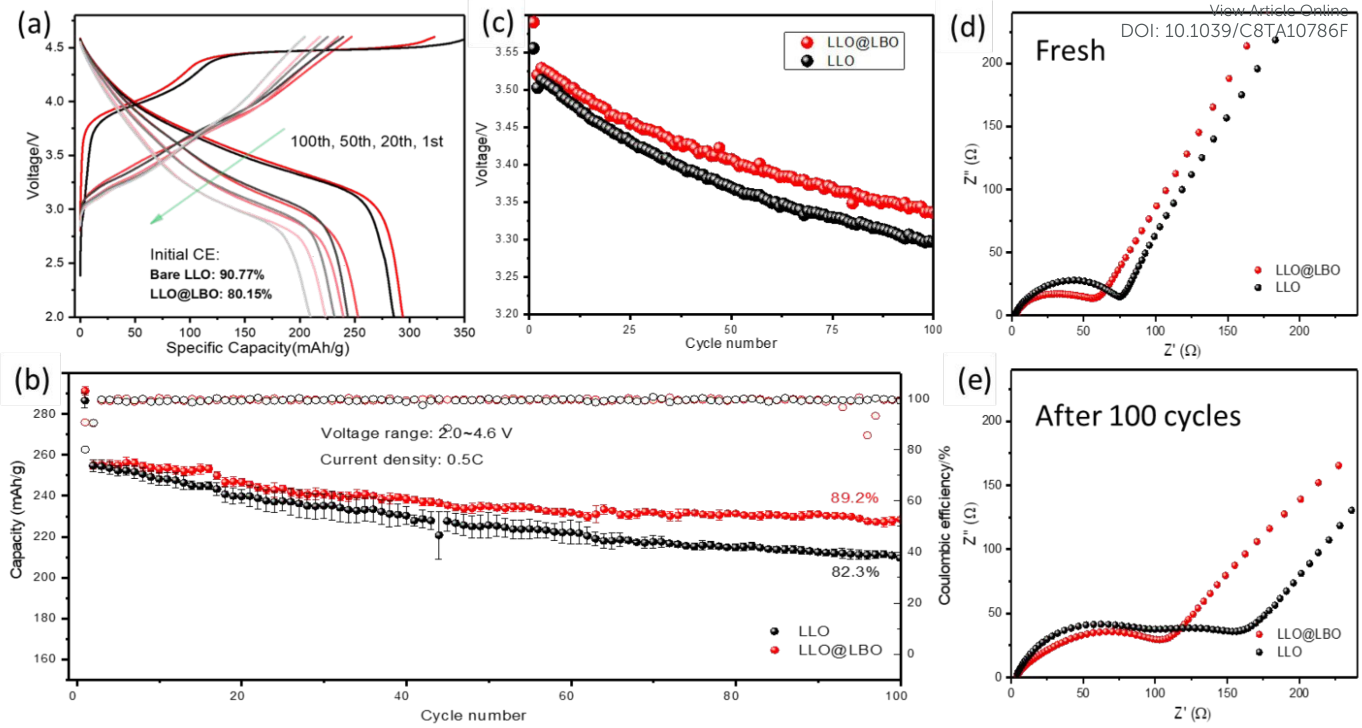

Fig. 3 Electrochemical performance of LLO and LLO@LBO. (a) charge/discharge profile at 1st, 20th, 50th and 100th cycle, (b) cycle performance and coulombic efficiency comparison of LLO@LBO and (c) average discharge potential versus cycle number plots. EIS spectra of LLO electrode and LLO@LBO electrode (d) before cycling and (e) after 100 cycles.

The electrochemical charge/discharge measurements of LLO and LLO@LBO were carried out using Li metal as the counter electrode between 2.0 and 4.6 V at constant current 0.5 C (1 C = 200 mA g^{-1}). The initial charge curves of LLO and LLO@LBO are composed of a slope region below 4.5 V, followed by a voltage plateau at around 4.5 V (Figure 3a). The slope region results from the oxidation of TM (Co^{3+} and Ni^{2+}) and the voltage plateau is related to activation of Li_2MnO_3 phase and oxygen redox, which is close to the extra specific capacity for LLOs cathode materials.[11] For LLO and LLO@LBO, the specific capacities of the slope region are similar. But the voltage plateau of LLO@LBO is a little shorter than that of LLO. This is due to supressed side reactions resulting from the strong B-O bond which stabilizes the oxygen framework of LLO@LBO.[29] The mass of introduced boron atoms is only 0.16 wt% of the host, far lower than the LLOs with other multivalent ions doping (Table S2). As a result, LLO@LBO shows a lower initial charge capacity (322.6 mA h g^{-1}) but a higher discharge capacity (293.9 mA h g^{-1}) compared to those of LLO (355.4 and 285.6 mA h g^{-1}, respectively). Initial Coulombic efficiency increases from 80.15% to 90.77% after boron doping. The enhanced reversible capacity and initial Coulombic efficiency is superior to most of previous work (Table S2). Figure 3b shows the cycling performance of the cells at 0.5 C rate. LLO@LBO shows obviously enhanced cycle stability. A much larger discharge capacity of 228.55 mA h g^{-1} was obtained after 100 cycles, corresponding to a capacity retention of 89.5%. On the contrary, the discharge capacity of LLO decreases from 254.72 mA h g^{-1} to 209.6 mA h g^{-1} after 100 cycles with a capacity retention of 82.3%. To estimate the voltage fading, the average voltage is calculated based on cycling data at room temperature (Figure 3c). The average voltage of LLO@LBO decreases from 3.5291 V to 3.3355 V after 100 cycles, 1.936 mV per cycle, significantly decreased from 2.163 mV for the case of LLO. The reasons of suppressed voltage fading will be discussed in next section.

Electrochemical impedance spectra (EIS) were measured to explore the origin of the improvement in the electrochemical performance of LLO@LBO. Nyquist plots of LLO and LLO@LBO electrodes are compared before charging (Figure 3d) and after 100 cycles at 0.5 C (Figure 3e). The plots of LLO and LLO@LBO before charging consist of a semicircle in intermediate-frequency region and a sloped straight line in the low-frequency region. The semicircle in the intermediate-frequency region is correlated with the charge-transfer (R_{ct}) process, and the straight line in the low-frequency region is related to the solid-state diffusion (Warburg impedance, Z_w) of Li ions in the active materials.[6,20] When there are two semicircles in the Nyquist plot in this case, the semicircle at high frequency belongs to the surface layer resistance (R_{sl}) resulting from solid electrolyte interface (SEI) while the other semicircle at low frequency is related to R_{ct}. The equivalent circuits are shown in Figure S8 and the simulated results are shown in Table S3. Before charging, the R_{ct} value of LLO electrode is 79.55 Ω while the LLO@LBO electrode exhibits smaller R_{ct} value of 56.64 Ω. The diffusion coefficient of Li^+ has been calculated in the Figure S9 based on the EIS results. After boron doping, the diffusion coefficient of Li^+ increased from 4.27x10^{-16} to 1.18x10^{-15} cm^2 s^{-1}. After 100 cycles, while R_{ct} values of both samples increase because the structural integrity would be compromised inevitably,[40] but the R_{ct} value of LLO@LBO electrode only increases to 131.6 Ω compared to 209.44 Ω for the undoped LLO case. This indicates that LLO@LBO electrode has a more stable structure benefiting

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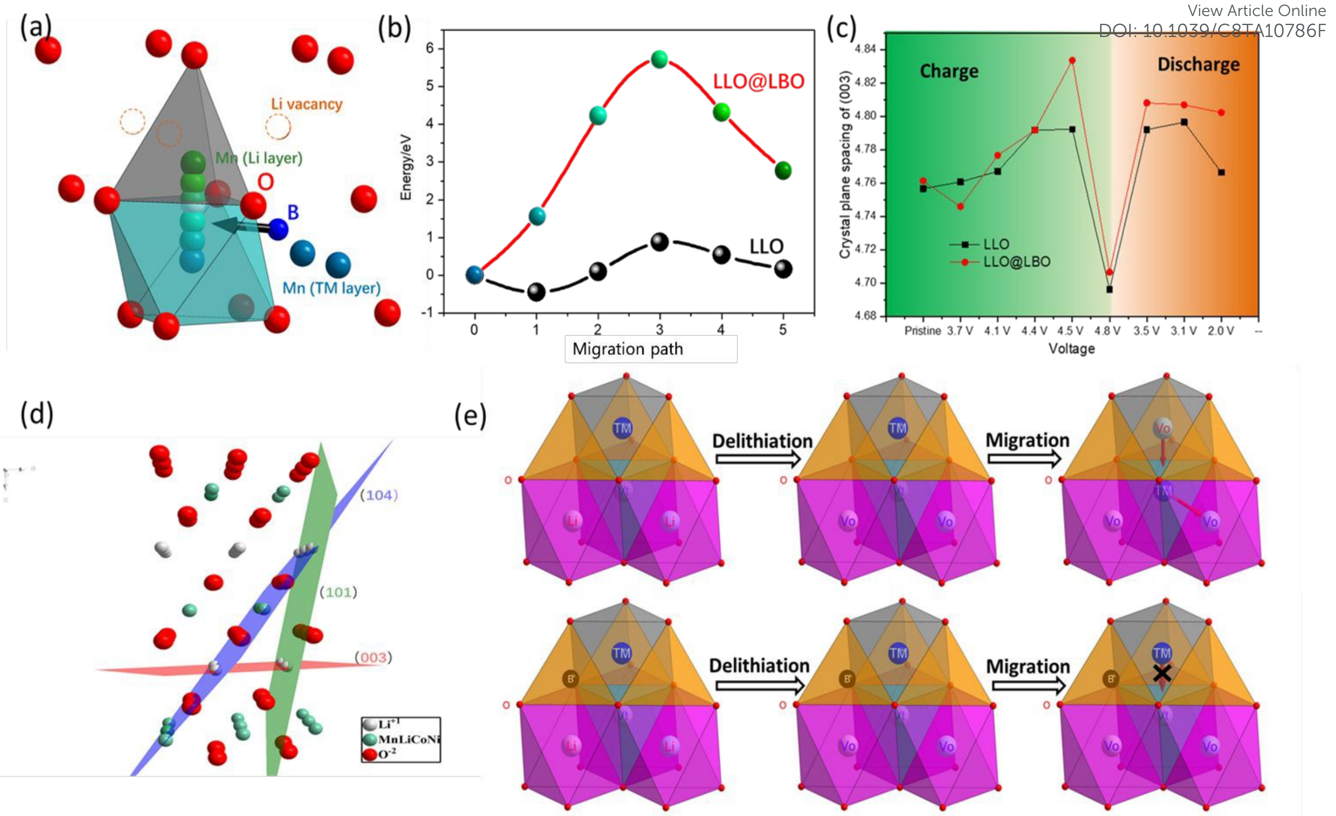

Fig. 4 (a) The migration module of Mn in LLO@LBO. (b) The energy of Mn at different positions in LLO@LBO and LLO. (c) Crystal plane spacing of (003) at different charge/discharge states. (d) Illustration of the atomic configure and reflection planes of (104), (003) and (101) in the layered structure. (e) Schematic diagrams of the phase evolution routes for LLO and LLO@LBO and it was built on O3 layered model with unrealistic radii ratios, V_o and V_t stands for octahedral vacancy and tetrahedral vacancy, respectively.

from boron doping, and leads to a relatively low R_{ct} value and enhanced electrochemical performance.[40] Furthermore, after 100 cycles the Nyquist plots of LLO and LLO@LBO shows distinct difference. Extra semicircle at high-frequency parts are observed in Nyquist plots of LLO (Figure 3d). But the Nyquist plot of LLO@LBO after 100 cycles remains similar shape as that before cycling (Figure 3d). The appearance of second semicircle (corresponding to R_{sl} =116.97 Ω) of LLO electrode after 100 cycles suggests an unstable surface structure. It could be related to the oxygen loss during cycling. In the case of LLO@LBO electrode, the initial pattern of EIS remains unchanged, indicating a much improved structure stability.[40]

The thermal stability of cathode materials is closely related to the battery safety. Therefore, the differential scanning calorimetry (DSC) measurements of LLO and LLO@LBO electrodes were carried out at the fully charged state of 4.6 V without further treatment. The obtained DSC profiles are presented in Figure S10. For LLO, the exothermic reaction initiated at around 218 ℃, but it was delayed to 231 ℃ with an obvious decrease in heat release (628.08 and 192.02 J g^{-1} for LLO and LLO@LBO, respectively) after boron doping. It is expected that the existence of B-O bonds with high covalency would stabilize the oxygen framework and enhance the battery safety.[25]

The mechanism for the improved electrochemical performance of LLO@LBO

To investigate the effect of boron doping in detail, we conducted the first-principle calculation to study the process of TM migration during initial charging process. Figure 4a shows the migration path of TM, from octahedral sites at the TM layer to tetrahedral sites at the lithium layer.8 Considering the high Mn content in LLO, we focus on the migration process of Mn to understand TM migration. We choose six sites from the migration path and calculated the energy of Mn ions at each site (Figure 4b). The energy of Mn ions in LLOs went down at beginning of the migration process, which means that Li ion extraction accelerates Mn migration. In contrast, the energy decrease of Mn ions at beginning of migration has not been observed in LLO@LBO. This indicates that the introduction of boron could enhance the structural stability of LLO@LBO, consistent with the results of EIS (Figure 3d and e) and DSC (Figure S10, ESI†). Importantly, the energy barriers of Mn ion migration in LLO@LBO is six times of that for the LLO case, indicating that the introduction of boron impedes the Mn migration process significantly, though the two cases have similar Mn ion migration path.

Ex suit XRD at different charge of state were conducted to study the variation of crystal structure. The XRD patterns (Figure 1) of LLO show three major peaks, (003), (101) and (104),[35] and their positions are labelled with different colours in the Figure 4d. Among them, (003) lattice plane is closely related to the electrochemical performance because the transportation of Li ions is along this plane.[10] Therefore, we've focused on the (003)

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lattice plane spacing variation at different charge of state to understand the effect of boron doping (Figure 4c). As the cell being charged, the (003) lattice plane spacing increases gradually. It is contributed to the increasing electrostatic repulsion between oxygen slabs due to the extraction of Li^+ from the Li layers.[41] When charged beyond 4.4 V, the lattice plane spacing of LLO@LBO continually increases while that of LLO shows no further increase. Previous works have proved that activation of Li_2MnO_3 phase is accompanied by the TM migration, which occurs at the special voltage plateau (around 4.5 V).[42] Thus, we believe the migration of TM ions from TM layers to Li layers would reduce the (003) lattice plane spacing due to the electrostatic attraction of the cations and anions. This would neutralize the effect of the extraction of Li^+ from the Li layers and results in unchanged lattice plane spacing. For our B doped material LLO@LBO, the continually increased lattice spacing indicates the Mn ion migration is mitigated. This agrees with the results of the first-principle calculation shown in Figure 4b. When charged to 4.8 V, the (003) lattice plane spacing shows a sudden decrease for both LLO and LLO@LBO. This is consistent with the previous work,[43] which results from the deconstruction of the framework of the lattice structure. At the following discharging process, the (003) lattice plane spacing rises back and then decreases slowly due to the Li ion reinsertion. Note that the (003) lattice plane spacing at the end of discharging process is larger than the pristine value for both LLO and LLO@LBO. This could be caused by the increasing electrostatic repulsion between oxygen slabs due to irreversible lithium loss from lithium layer. Nevertheless, the (003) lattice plane spacing of LLO@LBO is still larger than that of LLOs during discharging process due to the difference of TM migration in these two cases.

Based on the results of the first-principle calculation (Figure 4b) and ex suit XRD (Figure 4c), the schematic diagram is shown in Figure 4e to understand the effect of boron doping on TM ion migration. Voltage fading causes continuous energy density decay during cycling and limits the performance of LLOs seriously.[44] It is usually ascribed to the irreversible phase transformation from layered structure to spinel structure induced by the migration of TM.[10] During charging process, the Li^+ ions extract from LLOs cathode electrode gradually, and Li vacancy increment accelerates the TM migration from octahedral sites of TM layer to tetrahedral sites of lithium layer and then to octahedral sites of lithium layer. In the following discharging process, lithium ions reinsert into the lattice and occupy the tetrahedral sites at Li layer due to the effect of migrated Mn ions,[45] resulting in the formation of spinel structure and voltage fading.[42] For LLO@LBO, boron atoms are doped into the tetrahedral sites of TM layer and hinder the migration of TM greatly due to the strong repulsion between B^{3+} and migrated TM ions. Thus, the voltage and capacity decay are mitigated greatly.

Full cell performance study

While the LLO@LBO material exhibits a significantly improved performance in the half-cell device, it is important to

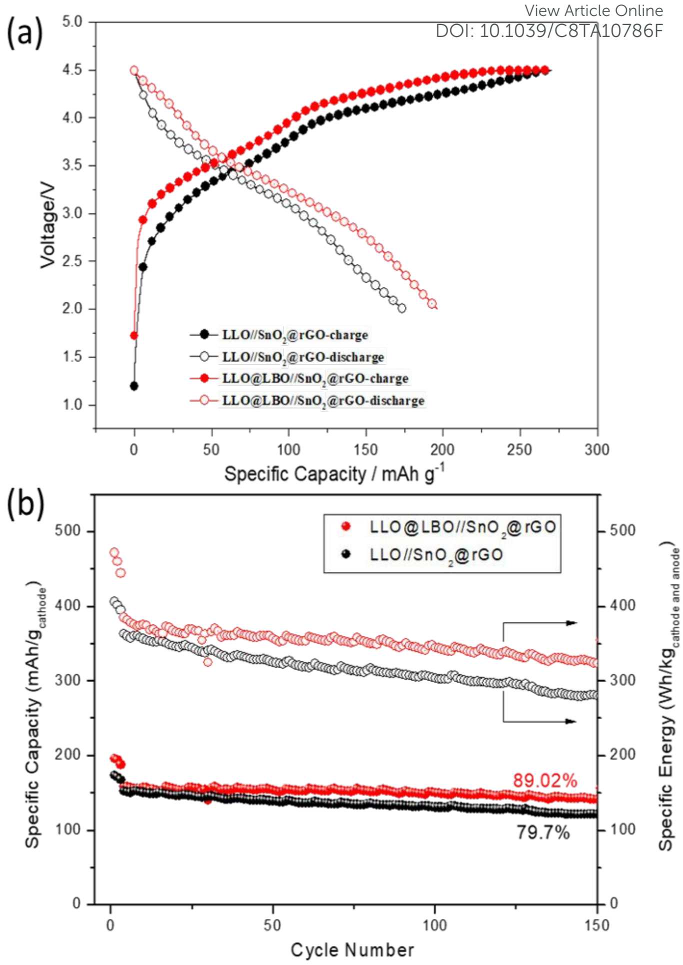

Fig. 5 (a) charge/discharge profile at initial cycle of full cells LLO//SnO2@rGO and LLO@LBO//SnO2@rGO. (b) The cycle performance and energy retention of LLO@LBO//SnO2@rGO and LLO//SnO2@rGO.

have more reliable evaluation using full cell device. Obviously, for cycling performance evaluation of this material, the anode material needs to have long cycling time. Also, considering the high energy density of LLO@LBO, similar or higher energy density is required for this anode material. With these, anode material SnO_2 was used as it exhibits both high energy density and long cycling time,[46] with combination of reduced graphene oxide (rGO) to compromise the volume change during the cycling of SnO_2. Its SEM images is shown in Figure S11a and the SnO_2 particles are distributed uniformly in the rGO host. Electrochemical performance test shows that SnO_2@rGO anode materials deliver an initial CE of 81% and an stable specific capacity of around 750 mA h g^{-1} at 0.5 A/g (Figure S11b and c, ESI †). Thus, LLO@LBO was coupled with SnO_2@rGO as the anode material in a lithium ion full cell to investigate LLO@LBO electrochemical performance. Furthermore, Sn element can be used as an internal standard substance to reduce the experimental error for ICP (Inductively Coupled Plasma) analysis of Mn content.

Figure 5a shows the initial charge/discharge profiles of the full cell LLO//SnO_2@rGO and LLO@LBO//SnO_2@rGO. The reversible specific capacity of LLO@LBO//SnO_2@rGO is 196.4 mA h g^{-1} based on the mass of cathode materials, much larger than that of 173.8 mA h g^{-1} for the full cell LLO//SnO_2@rGO and

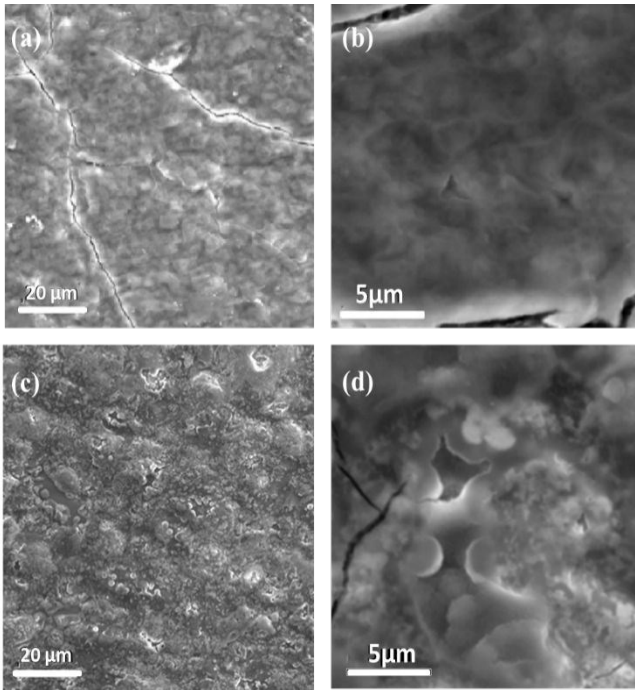

Fig. 6 The morphology of anode electrodes after 20 cycles. (a) LLO@LBO//SnO2@rGO and (c) LLO//SnO2@rGO and their enlarged images (b) and (b), respectively.

indicating the enhanced specific capacity due to the boron doping. As a result, the initial Coulombic efficiency increased from 64.2% to 73.7%. Figure 5b shows the cycle performance of full cell LLO//SnO_2@rGO and LLO@LBO//SnO_2@rGO. The discharge energy density of LLO@LBO//SnO_2@rGO is as high as 472.1 Wh Kg^{-1} at 0.05 C based on the total mass of anode and cathode materials. Moreover, it still keeps an energy density of 323.9 Wh Kg^{-1} after 150 cycles at 0.5 C with an energy retention of 84.1%. In contrast, the undoped LLO//SnO_2@rGO only displays an energy density of 406.3 Wh Kg^{-1} at 0.05 C and 280.3 Wh Kg^{-1} after 150 cycles at 0.5 C with an energy retention of 77.2 %. To explore the effect of boron doping on anode electrodes, the morphology of cycled anode electrode was characterized by SEM (scanning electron microscopy) (Figure 6). After 20 time cycling, the surface of anode electrode in LLO@LBO//SnO_2@rGO is still smooth (Figure 6a and 6b), indicating an stable SEI structure. In contrast, the surface of that in LLO//SnO_2@rGO became rough and filled with embossments and hollows (Figure 6c and d). Previous reports have indicated that the surface roughness and erosion during the cycling is due to the reduction of Mn^{2+}, which cause Mn metal deposition on the surface of the anode electrode.[47-49] Furthermore, since Mn metal has higher reactivity than graphite materials, the deposited Mn could catalyse and cause the solvent decomposition, further limiting the cycling time.[50,51] Therefore, we believe that the large difference of the morphology of LLO and LLO@LBO electrodes result from the different level of Mn^{2+} dissolution of the cathode electrode. Based on this, the ICP experiment of the cycled anode electrodes was carried out to detect the Mn content. The ICP (Table 1) demonstrates that the Mn/Sn ratio (Sn as internal standard substance) of the anode electrode decreases from 0.195 to 0.070 after boron doping, indicating the enhanced structure stability of the doped LLO@LBO.

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Table 1 The ICP results about the content of Mn and Sn in cycled anode electrode of full cell LLO//SnO_2@rGO and LLO@LBO//SnO_2@rGO

	Mn(mmol/L)	Sn(mmol/L)	Mn/Sn
LLO//**SnO_2@rGO**	0.024	0.123	0.195
LLO@LBO//**SnO_2@rGO**	0.006	0.086	0.070

Conclusions

We have demonstrated that boron doping LLOs shows improved discharge capacity and cycle performance, suppressed migration and dissolution of Mn ions and enhanced structural stability. Firstly, the strong B-O bonds can stabilize the oxygen framework of LLOs and improve the structural stability, which can reduce oxygen loss and supress the side reactions, resulting in improved initial Coulombic efficiency. Secondly, the boron atoms are occupying the tetrahedral sites at TM layer and enlarging the migration energy barrier of TM ions during charging process effectively. This should play a key role in inhabiting voltage fading. Both of them are critical factors of the cycle performance improvement of LLO@LBO in half cell. As a result, the doped material LLO@LBO delivers a reversible capacity of 293.9 mA h g^{-1} with a capacity retention of 89.5% at 0.5 C after 100 cycles in half cell. Moreover, the full cell LLO@LBO//SnO_2@rGO also shows improved reversible energy density and energy retention compared to LLO//SnO_2@rGO. Except the superior cycle performance of the cathode and anode materials, the results of ICP and SEM demonstrated that the Mn ion dissolution is supressed after boron doping. The full cell LLO@LBO//SnO_2@rGO shows a high initial energy density (472.1 Wh kg^{-1}) with excellent energy retention (84.1% retention after 150 cycles at 0.5 C). The studies of the effect of boron doping on LLOs in half and full cells are crucial to further explore the Li-ion batteries with higher energy density and longer cycle life which can meet the ever-growing safety requirement.

Experimental section

Synthesis of $Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}O_2$ (LLO) and boron doping $Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}O_2$ (LLO@LBO)

Li-rich layered oxide, $Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}O_2$, was synthesized by oxalic acid co-precipitation, hydrothermal and calcination processes modified based on ref 53. Firstly, the stoichiometric amount of $CH_3COOLi\cdot 2H_2O$ (10% excess), $Ni(CH_3COO)_2\cdot 4H_2O$, $Co(CH_3COO)_2\cdot 4H_2O$ and $Mn(CH_3COO)_2\cdot 4H_2O$ were dissolved in the ethanol solution together, and the resulting solution is continuously stirred at room temperature for 0.5 h. Oxalic acid was added drop by drop into the mixed ethanol solution as a precipitating agent. The obtained mixture was then transferred into a sealed PTFE (Polytetrafluoroethylene) container followed by heating at 180 ℃ for 12 h. After the hydrothermal reaction, ethanol solvent was removed by vacuum filtration and dried at 120 ℃ for 3 h. The solid mixture was then pressed into a plate and preheated in air at 450 ℃ for 5 h and then calcined in air at 850 ℃ for 15 h. The final product was obtained after uniform grinding. $Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}O_2$ was mixed with $LiBO_2$ by ball milling (500 rpm and 1h). Afterwards, the solid mixture was pressed into a plate under a pressure of 10 MPa. The plates were then calcined in flowing argon at 700 ℃ for 2 h. The final product was obtained after uniform grinding.

Characterization

Power X-ray diffraction (XRD) analysis was performed on a Rigaku D/Max-2500 diffractometer with Cu Kα radiation. As for ex suit XRD characterization, the samples were prepared by charging (discharging) the cells at constant voltage and then disassembled in an argon-filled glove box. The cathode electrode was sealed in polyimide film to avoid the reaction with air. DSC measurement (Netzsch-STA 449C) was conducted from room temperature to 300 ℃ at a heating rate of 10 ℃ min^{-1}. Samples were prepared by charging the cells at 4.6 V and maintained at a constant voltage for 10 h, and then disassembled in an argon-filled glove box. The cathode material was sealed in an aluminium DSC pan with an additional 1 μL new electrolyte before taken out from glove box for the measurement. The morphological characteristics were evaluated using scanning electron microscopy (SEM, PhenomPro) and transmission electron microscopy (TEM, JEM-2800) The content of Mn and Sn was detected by inductively coupled plasma-optical emission spectroscopy (ICP-OES, Perkin Elmer, Optima 5300DV) after microwave digestion of cycled anode electrode. X-ray photoelectron spectroscopy (XPS) was carried out using Escalab 250Xi (Thermo Scientific).

The calculations have been performed using the ab initio total energy and molecular dynamics program VASP (Vienna ab initio simulation program). Total energy calculations based on density functional theory (DFT) were performed with the exchange and correlation functional form. The interaction of core electrons with the nuclei is described by the projector augmented wave (PAW) method.[54] A plane wave cutoff energy of 300 eV was used for all calculations. The integration in the Brillouin zone is done on an appropriate k-point set determined by the Monkhorst-Pack scheme. Geometry relaxation of atomic positions with the conjugate gradient algorithm was used to obtain the local minima of the systems. The stopping criteria used were energy differences less than 0.1 meV and forces less than 0.01 eV/Å. We used the Nuged Elastic Band method (NEB) implemented in VASP to investigate whether the addition of B in $Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}O_2$ inhibits the migration of lithium ions. Large supercells are required to attain the dilute limit, but since NEB calculations are expensive, the size of the supercell is limited by the computational resources. In the present work, we used Li_3MnO_6 supercells (2 × 2 × 1) with a 2 x 2 x 1 k-mesh. Constant volume calculations were performed for four intermediate images initialized by linear interpolation between the two fully relaxed end points.

Cell fabrication and electrochemical characterization

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Electrochemical measurements were carried out using coin-type cells. To prepare working electrodes, LLO@LBO composite, Super P carbon black, and poly(vinylidene fluoride) (PVDF, in N-methyl-2-pyrrolidone) with mass ratio of 80: 10: 10 were mixed to produce a homogeneous slurry and coated onto aluminium foil. After heating at 60 ℃ for 3 h and 150 ℃ for 1 h under vacuum, the electrode sheet was pressed and punched into ~12 mm diameter electrodes with a mass loading of ~2.5 mg. The CR2430 coin-type cells were assembled in an argon-filled glove box with lithium metal foil as the counter/reference electrode. CR2430 coin-type full cells with the LLO@LBO (LLO) as the cathode materials and SnO_2@rGO as the anode materials were also assembled in the high-purity argon-filled glove box. The tested anodes were prepared using similar methods at that of the LLO@LBO electrodes. The resultant anode slurry was dispersed and spread onto the copper foil. Before assembling the full cell, the anode electrodes are treated by a pre-lithiated process with 20 cycles to form stable SEI and stopped with a delithiated state. The ratio of $C_{anode}/C_{cathode}$ is about 1.1 to make full use of the capacity of cathode electrode. The electrolyte (www.dodochem.com) was 1 M LiPF6 in ethylene carbonate (EC): dimethyl carbonate (DMC): diethyl carbonate (DEC) (1:1:1 vol%). Clegard 2325 was used as the separator. The charge/discharge measurements were performed using a battery test system (LAND CT2001A model, Wuhan LAND Electronics. Ltd.) over a voltage window from 2.0 V to 4.6 V at room temperature. For full cell test, the voltage window of 2.0~4.5 V is adopted for activated and following cycle process. Serval low rate (0.05 C) cycles at beginning are always necessary to activate the Li_2MnO_3 phase. Electrochemical impedance spectral (EIS) measurements were recorded using an Autolab system (Metrohm) and carried out at AC amplitude of 10mV in the range of 100 kHz- 10 mHz.

Conflicts of interest

There are no conflicts to declare.

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